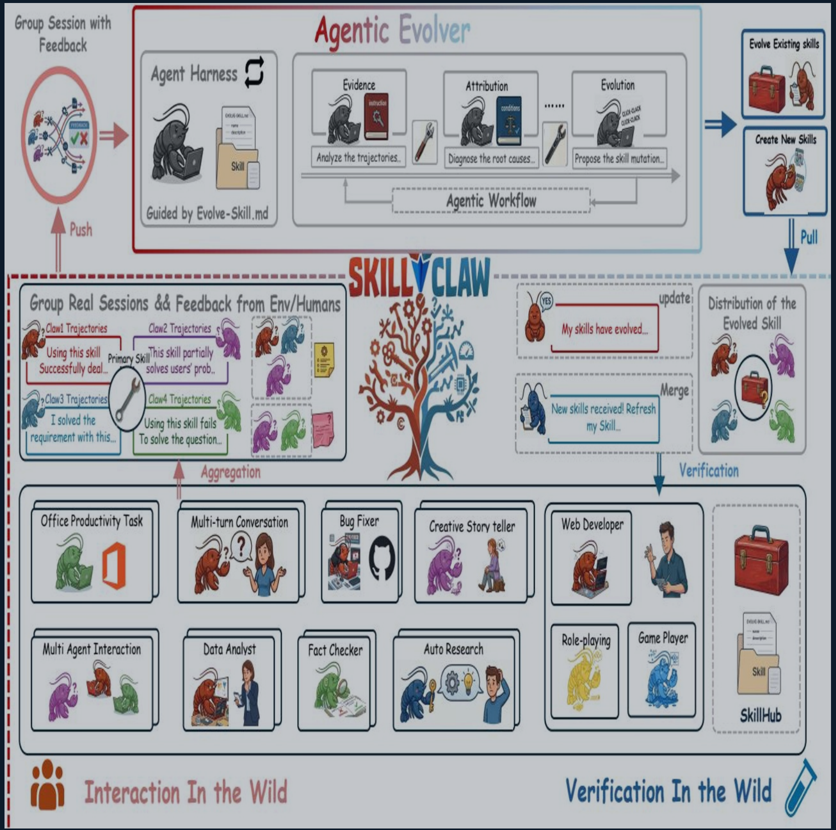
SkillClaw

Collective Skill Evolution for Multi-User LLM Agent Ecosystems

Skills that learn from every user. Automatically.

ML Researchers · AI Systems Engineers



arxiv.org/abs/2604.08377

Agent Skills Remain Static After Deployment

Users Rediscover Solutions Independently

STATIC SKILL ECOSYSTEM

- Skills manually installed from hub
- No updates from runtime experience
- Failures repeat across users
- Each user rediscovers same solutions
- Knowledge never accumulates at system level

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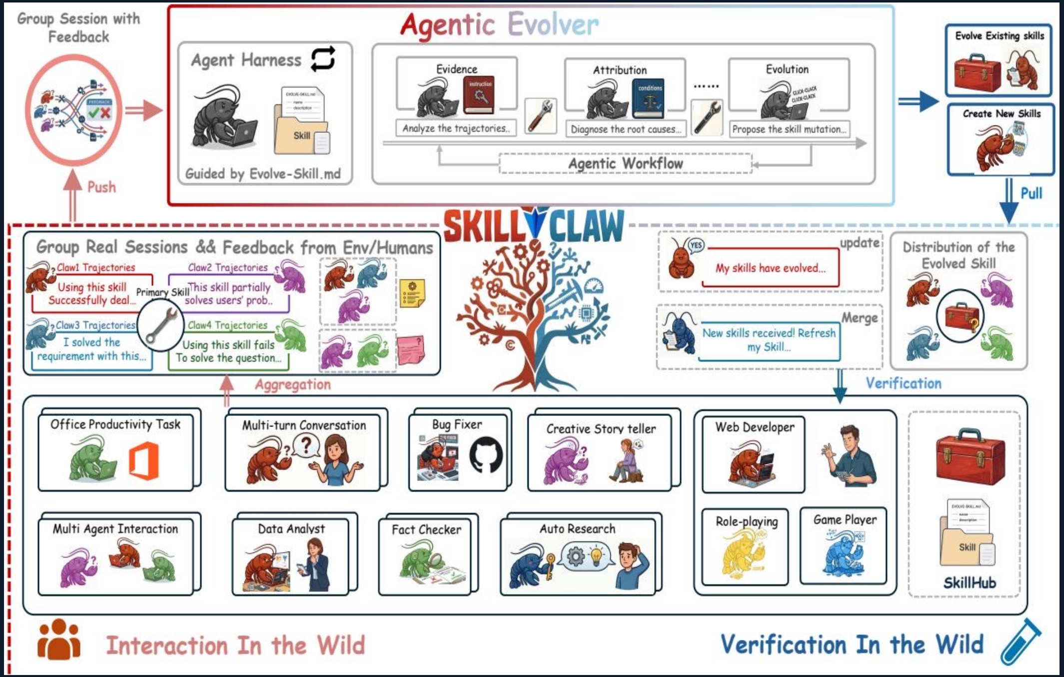
EVOLVING SKILL ECOSYSTEM

- Skills evolve from runtime interactions
- Cross-user knowledge aggregation
- Patterns of failure → system improvements
- Solutions shared across the ecosystem
- Continuous, automatic skill refinement

Memory methods store instances but don't generalize. Skill libraries compress experience but remain static.

Closed-Loop Pipeline: Multi-User Interaction → Shared Skill Evolution



1 EVIDENCE

Analyze trajectories
Identify behavioral patterns

2 ATTRIBUTION

Diagnose root causes
of failures & successes

3 EVOLUTION

Propose skill mutations
Refine or create skills

4 DISTRIBUTION

Verify & merge updates
Sync back to all agents

CLOSED LOOP — CONTINUOUS IMPROVEMENT

Evidence → Attribution → Evolution in Three Stages

Open-ended agent reasoning — not predefined update rules

01 EVIDENCE

Trajectory Analysis

- Aggregate structured session trajectories
- Group trajectories by referenced skill
- Preserve full action–feedback causal chains
- Identify recurring patterns of success & failure

02 ATTRIBUTION

Root Cause Diagnosis

- Diagnose why failures occur for specific skills
- Distinguish generalizable issues from edge cases
- Leverage cross-user diversity for robust signal
- Single-user data insufficient for generalizable fix

03 EVOLUTION

Skill Mutation Proposals

- Propose targeted skill mutations via open reasoning
- Refine existing skills or synthesize new ones
- Verify candidates before merging to shared repo
- Reject regressions — conservative by design

The evolver performs open-ended reasoning over interaction evidence — no hard-coded update logic.

Cross-User Trajectories Expose Skill Behavioral Boundaries

A single user rarely generates enough signal to separate a generalizable improvement from an idiosyncratic fix.

Compatible Claw-style systems: OpenClaw · CoPaw · IronClaw · PicoClaw · ZeroClaw · NanoClaw · NemoClaw

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Diverse Contexts

Different users exercise the same skill under varied conditions, producing complementary behavioral views.

⨁

Complementary Signals

Reveals both conditions under which a skill works and those under which it breaks — success AND failure modes.

★

Generalizable Improvements

Aggregated cross-user evidence enables the evolver to propose updates that hold beyond any single user's context.

Multimodal Creative Pipeline Skill — 6 Days of Evolution

Conservative verification: only 1 of 6 candidates accepted

Purpose: prevents regressions in multi-user shared skill ecosystem

Day	Candidate Skill	Skill Function	Verdict	Next-Day Action
1	validate-tmp-workspace-inputs	Check /tmp_workspace inputs & environment setup	**ACCEPT**	Upgrade to new best pool
2	multimodal-input-validation-and-setup	Multimodal input validation & output env init	**REJECT**	Keep current best pool
3	multimodal-creative-task-pipeline	Multimodal creative pipeline (extract from PDF/video/image)	**REJECT**	Keep current best pool
4	multimodal-creative-task-pipeline (impr.)	Added image classification, visual generation, structured validation	**REJECT**	Keep current best pool
5	...-pipeline (impr.) + validate-required-input-files	Creative pipeline + per-file fail-fast validation	**REJECT**	Keep current best pool
6	multimodal-creative-task-pipeline (cand.)	Extended PDF-to-poster / document-to-visual paths	**REJECT**	Not admitted to next cycle

Git Push Auth-Fallback Skill — Iterative Refinement Over 6 Days

Iterative refinement: 3 of 6 candidates accepted before plateau

Contrast: more evidence → more accepted changes than conservative case

Day	Candidate Skill	Change Summary	Verdict	Next-Day Action
1	git-push-with-auth-fallback	Safe fallback instead of blocking on push failure	**ACCEPT**	Add to Safety best pool
2	git-push-with-auth-fallback	Unified patch/bundle filenames, reduced inconsistency	**ACCEPT**	Keep updated best pool
3	git-push-with-auth-fallback + git-clone-to-directory	Auth-alternative paths & secrets audit; fixed subshell pitfalls	**ACCEPT**	Keep current best pool
4	(none)	Same-pool retest; validator confirmed current pool as best	**ACCEPT**	Continue same best pool
5	git-push-with-auth-fallback	Added "push hang treated as auth failure"; no improvement	**REJECT**	Keep current best pool
6	git-push-with-auth-fallback	Added identity config & filename consistency; no improvement	**REJECT**	Not admitted to next cycle

Three Properties That Distinguish SkillClaw

01

Collective Evolution

Individual user interactions contribute to a shared, continuously improving skill ecosystem. Knowledge accumulates at the system level — not isolated per session.

02

Fully Automatic

Skill evolution is driven entirely by runtime interaction without manual curation or explicit user intervention. No human in the loop required.

03

Agentic Evolution Paradigm

Skill updates are produced through open-ended reasoning over interaction evidence — not predefined rules. The evolver acts as an autonomous reasoning agent.

Implications

Continuously improving agent systems require collective, cross-user evidence — not just per-session memory. Evaluation on WildClawBench (Qwen3-Max backbone) demonstrates substantial improvements across tasks.